

Stage-Aware Early Warning of Australian Winter Crop Yield Shortfall Using Daily Weather and Soil Data

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Abstract

Early warning of crop yield shortfall is useful only when risk can be updated before the season is effectively complete and when forecast skill is not driven by post-harvest leakage. We study Australian winter crops at state level using 966 region-crop-year yield observations from 1989–2021, daily SILO weather, and Soil and Landscape Grid attributes. Daily weather is converted into May-Jun, May-Jul, May-Aug, May-Sep, and May-Oct feature windows. We separate a history-free weather-soil setting from a history-enhanced operational setting to distinguish within-season environmental signal from persistent crop-region yield memory. On held-out 2017–2021 seasons, the best history-free May-Oct model reaches RMSE 0.889 t/ha and R^2 0.507, while the best operational May-Oct model reaches RMSE 0.658 t/ha and R^2 0.730. Ablation and historical baselines show that weather and soil provide measurable but secondary monitoring information; persistent crop-region structure and lagged yield history account for much of the strongest operational skill. Risk classification and uncertainty diagnostics support the framework as a state-level monitoring tool for known crop-region histories, while clearly limiting causal, farm-level, and unseen-region claims.

Keywords: crop yield forecasting; early warning; climate risk; Australian agriculture; weather features; soil data; uncertainty

1. Introduction

Australian dryland grain systems face recurring climate risks, including rainfall deficits, heat events, frost, and periods of high evaporative demand. These risks are particularly important for winter crops, where the difference between early vegetative conditions and late-season grain-filling conditions can change the decision value of a forecast. Prior work has shown that Australian wheat yields have been affected by adverse climate trends and by drought-heat exposure [9, 5]. For decision support, however, the operational question is not only whether yield can be estimated after the season, but whether risk can be updated at meaningful within-season stages.

This paper studies state-level early warning for Australian winter crops. The task is to predict yield and low-yield shortfall risk for each region, crop, harvest year, and forecast window. Unlike a post-hoc anomaly attribution or multi-method XAI study, the central question is operational timing: how early can yield-shortfall risk be monitored? A May-Jun estimate supports early monitoring, while a May-Oct estimate approaches a full-season pre-harvest assessment. This framing is aligned with recent early-warning work that emphasizes anticipatory crop-failure monitoring rather than post-season diagnosis alone [1].

The empirical design is intentionally conservative. We remove production and area variables from the main predictors, compute expected-yield and weather-deviation baselines

from training years only, and keep the 2017–2021 test period outside all scaling, thresholding, calibration, and residual estimation steps. This avoids turning post-harvest information into apparent forecast skill.

The main contribution is a stage-aware framework that separates two claims that are often mixed. First, a history-free weather-soil model tests whether daily weather, train-derived weather deviations, and soil background provide measurable evidence of within-season monitoring signal. Second, an operational model adds lagged yield history to quantify how much forecast quality improves when historical production memory is allowed. This distinction matters because a strong operational forecast may partly reflect persistent crop-region productivity rather than purely within-season weather signal.

We make three contributions. First, we formulate Australian winter-crop forecasting as a lead-time-aware shortfall-monitoring problem, where predictions are updated from May-Jun to May-Oct rather than evaluated only after the season is nearly complete. Second, we separate a history-free weather-soil setting from a history-enhanced operational setting, allowing weather-derived early-warning evidence to be distinguished from persistent crop-region productivity memory. Third, we evaluate practical usefulness through temporal holdout testing, feature-regime ablation, stress validation, threshold-tuned risk screening, and uncertainty diagnostics, framing the outputs as state-level decision support rather than causal or farm-level prediction.

We organize the study around four research questions. RQ1 asks how much forecasting skill is available at successive monitoring windows from May-Jun to May-Oct. RQ2 asks how much of this skill comes from stage-aware weather and soil features compared with lagged yield history. RQ3 asks whether low-yield-risk predictions can support screening without becoming automatic decision triggers. RQ4 asks how robust the framework is under rolling-origin, leave-one-crop-out, and leave-one-region-out stress tests.

2. Related Work

2.1. Crop Yield Forecasting and Early Warning

Machine learning has become a common approach for regional crop-yield forecasting, with weather, soil, remote sensing, and management covariates appearing across many systems [24, 18]. At the same time, several studies warn that yield datasets are often small relative to the dimensionality of environmental predictors, making regularized and tree-based tabular models attractive for grain forecasting [15]. Recent work also pushes yield modeling toward multi-modal benchmarks and operational decision-support interfaces [13, 4]. Our setting is smaller and more targeted: state-level Australian winter crops with a fixed May-Oct monitoring calendar.

Early-warning studies differ from generic yield prediction because lead time is part of the objective. A model that works only after most weather information is observed may have limited preparedness value. Preseason and early-season crop-failure forecasts show the importance of evaluating when skill emerges, not only how accurate the final estimate is [1]. We adopt this view by reporting skill separately for each forecast window.

2.2. Stage-Aware Weather and Extreme-Event Features

Weather-yield relationships are stage dependent. Developmental-stage studies show that climatic means and weather extremes can have different effects depending on when they occur during crop growth [22]. Broader climate-risk evidence also emphasizes compound hot-dry extremes, crop-specific vulnerability, and heterogeneous regional responses [8, 23]. We therefore aggregate daily weather into stage-aware rainfall, heat, cold, evaporative-demand, radiation, and compound-stress features rather than relying only on full-season averages. Recent extreme-weather yield studies further motivate sparse or regularized modeling of drought, heat, flooding, and compound hazards [7, 11].

2.3. Interpretability, Uncertainty, and Decision Support

For agricultural risk monitoring, the output is used to prioritize attention rather than to prove a mechanistic causal pathway. Interpretability and uncertainty are therefore practical requirements, especially when models inform public or business decisions [21, 17]. We use transparent ablation, group permutation importance, and simple lead-time comparisons rather than making the paper a post-hoc XAI study. Explainable boosting and related glassbox tools provide one route for transparent tabular modeling [16], while conformal and conformalized quantile methods motivate distribution-aware intervals with explicit coverage diagnostics [20, 2].

3. Data and Prediction Targets

3.1. Study Panel and Data Sources

The analysis unit is a region-crop-year observation. The yield panel is derived from Australian Crop Report style state-level crop statistics [3]. It spans 1989–2021 and includes Barley, Canola, Lupins, Oats, and Wheat in New South Wales, Queensland, South Australia, Tasmania, Victoria, and Western Australia. Table 1 summarizes the panel. Each of the 966 yield observations is expanded over five forecast windows, giving 4,830 window rows. This expansion does not create new yield outcomes: models and metrics are evaluated within forecast windows, and all train/validation/test decisions are made by calendar year so the same region-crop-year outcome cannot be split across evaluation periods.

Table 1: Data summary for the Australian winter crop early-warning panel.

Item	Value
Analysis unit	State/region, crop, harvest year
Yield observations	966
Window-expanded panel rows	4,830
Years	1989–2021
Crops	Barley, Canola, Lupins, Oats, Wheat
Regions	6 Australian states/territories
Forecast windows	May-Jun, May-Jul, May-Aug, May-Sep, May-Oct
Split	train 701, validation 116, test 149

Daily weather comes from SILO-style Australian climate records, which are based on spatial interpolation of observed weather data [10]. We use daily rainfall, maximum and minimum temperature, radiation, evaporation, and vapor pressure to build cutoff-specific summaries. Soil variables are regional background attributes derived from the Soil and Landscape Grid of Australia [6]. They include topsoil and subsoil summaries of available water capacity, texture, bulk density, soil organic carbon, nutrients, pH, and cation exchange capacity, plus depth indicators. Because these soil summaries are region-level and static, they are treated as background vulnerability covariates, not as farm-level measurements or causal soil interventions.

3.2. Targets

The regression target is yield in tonnes per hectare, denoted $y_{r,c,t}$. We also define an expected-yield baseline from training years and the shortfall

$$s_{r,c,t} = \hat{y}_{r,c,t}^{\text{exp}} - y_{r,c,t}.$$

The expected-yield baseline is fitted using training years only. In the main specification, expected yield is estimated from a crop-region linear trend in calendar year; crop-level linear trends are used as fallback when a crop-region series is too short, and the overall training mean is the final fallback. This baseline excludes current-season weather, production, area, lagged yield, and all validation/test outcomes. Validation and test shortfalls are then computed by applying the fixed training-period baseline to later years. The binary low-yield-risk label is one when shortfall exceeds the crop-specific 80th percentile of the training distribution. We use this threshold as a broad watch-list definition rather than as a severe crop-failure definition. This target follows the general idea that yield anomalies or shortfalls require a reference trend before climate impacts can be assessed [14], but here the shortfall is used for early-warning classification rather than post-hoc causal attribution. The classification target is used for secondary risk-alert analyses; the main quantitative results remain regression and ablation metrics.

4. Stage-Aware Features and Models

4.1. Weather Windows

For each forecast window w , daily weather is aggregated only from May through the cutoff month of w . The design intentionally mimics an operational update: May-Jun uses only two months of weather, while May-Oct uses the longest pre-harvest window. No weather feature uses data after the cutoff month. This makes the lead-time comparison interpretable as a sequence of information sets rather than a single full-season model.

The weather feature groups are rainfall accumulation and dry spells, heat and cold exposure, radiation and evaporative demand, and compound hot-dry stress. Rainfall summaries include totals, rainy days, dry days, short-window maxima, heavy-rain days, and maximum consecutive dry days. Heat/cold features include mean and extreme maximum and minimum temperatures, heat-day thresholds, heat degree days, frost days, and cold days. Energy and dryness features include radiation, evaporation, and vapor pressure summaries.

Table 2: Feature groups used for stage-aware early-warning models.

Feature group	Examples	Interpretation
Rainfall/dryness	rain_sum, rain_days, dry_days, dry-spell length	Water supply and dry-spell stress
Heat/cold	tmax_mean, heat_days_30, heat degree days, frost days	Heat exposure and frost/cold risk
Energy/evaporation	radiation_sum, evap_sum, vapor pressure	Growth energy and evaporative demand
Compound stress	hot-dry days, high-evaporation dry days	Combined weather stress
Soil background	AWC, clay/sand, SOC, pH, bulk density, depth	Regional vulnerability conditioning

Compound features flag hot-dry days and high-evaporation dry days. These choices are motivated by stage-aware and extreme-weather yield studies [22, 8, 7].

We additionally construct weather-deviation features by subtracting train-period region-window baselines. This produces rainfall, temperature, dry-spell, heat-day, evapotranspiration, and radiation deviations that are comparable across regions and forecast windows. The weather-deviation baselines are computed using training years only, so they do not leak test-period information. In result tables, Dev denotes train-derived weather-deviation features.

4.2. History-Free and Operational Feature Sets

We report two primary feature regimes. The history-free weather-soil regime uses stage weather features, train-derived weather-deviation features, soil background features, crop identity, region identity, and year. It excludes lagged yield, rolling yield, production, and area. The operational regime adds past-yield features from prior years for the same crop-region series. This makes it useful for deployment in known crop-region histories, but it is not interpreted as pure weather evidence.

The model suite combines regularized linear baselines with gradient-boosted tabular models. Ridge and ElasticNet provide stable small-data baselines, while LightGBM and CatBoost provide nonlinear tabular benchmarks [12, 19]. We avoid presenting a black-box-only result: every performance table is paired with feature-set ablation, lead-time comparison, or group permutation importance. This is consistent with the broader argument that high-stakes decision support should prefer transparent or at least carefully audited modeling workflows [21].

Classification models use validation-tuned thresholds for low-yield-risk decisions. Instead of fixing a probability threshold at 0.5, we choose thresholds on validation years for F1, recall constraints, or precision constraints, then report test metrics once. Probabilistic interval diagnostics use quantile-style predictions and conformal calibration ideas as supplementary checks [20, 2]. These outputs are not framed as definitive event decisions; they are screening layers for state-level monitoring.

5. Evaluation Protocol

The main split trains on 1989–2012, tunes on 2013–2016, and tests on 2017–2021. The test period is not used for scalars, feature thresholds, residual calibration, conformal residuals, expected-yield baselines, weather-deviation baselines, or low-yield-risk thresholds. This temporal split reflects the intended forecasting direction: models learn from earlier seasons and are evaluated on later seasons.

Regression metrics are MAE, RMSE, and R2. Classification metrics are precision, recall, F1, ROC-AUC, PR-AUC, and Brier score. For probabilistic diagnostics, we summarize interval coverage and width. The conformal intervals target nominal 80% marginal coverage using validation-period calibration residuals only; test coverage is reported without retuning. Deviations from nominal coverage are interpreted as evidence of calibration difficulty under temporal shift and small calibration samples, not as a target corrected using test outcomes. The primary model-selection comparisons are always made within the same forecast window and split.

Robustness checks include rolling-origin folds, leave-one-crop-out validation, and leave-one-region-out validation. Rolling-origin validation tests sensitivity to a single time split. Leave-one-crop-out and leave-one-region-out are intentionally difficult transfer tests. We do not treat them as the main deployment setting, because the intended use case is monitoring crop-region combinations with historical observations. Instead, they identify where the framework is not yet robust enough for unseen-crop or unseen-state generalization.

6. Results

6.1. Lead-Time Skill

Table 3 compares the best history-free weather-soil and operational models at each lead time. For each window and feature regime, the displayed model is selected using validation years only, with RMSE as the selection metric. The operational model is already informative at May-Jun, with RMSE 0.721 t/ha and R2 0.676, and improves toward May-Oct. This early result is important because the May-Jun window contains only a partial season, yet still captures enough historical and early-weather structure for useful monitoring. The history-free model is weaker but still positive, reaching RMSE 0.889 t/ha and R2 0.507 at May-Oct. The history-free trajectory is not strictly monotonic: later windows add useful weather information but also correlated features and noise in a small state-level panel. We therefore interpret the curve as a monitoring trajectory rather than assuming that every additional month must improve skill. The gap between the two regimes motivates the central interpretation of the paper: weather and soil provide measurable but secondary monitoring signal, while historical yield memory is a major driver of operational accuracy. A fixed-model lead-time check in Appendix A verifies that the lead-time pattern is not solely an artifact of switching model classes across windows.

6.2. Ablation: Weather Signal Versus Yield History

Table 4 shows compact ablation results for the earliest and latest windows. The identity/time baseline is not trivial because crop and region encode persistent productivity differences. Weather-only and weather-plus-soil feature sets add stage-specific environmental information, while the full operational model combines this information with past-yield history. The history-free weather-soil models retain positive test skill, especially in the May-Oct window, but they do not consistently outperform identity/time or simple historical baselines. This limits the interpretation of the weather-soil setting: it provides evidence that within-season environmental features contain monitoring signal, not that weather and soil alone dominate persistent crop-region structure. Weather-deviation features are not

Table 3: Best test performance by forecast window for the history-free weather-soil model and the operational model. Dev denotes train-derived weather-deviation features.

Window	Feature set	Model	MAE	RMSE	R2
May-Jun	Weather+Dev+Soil	LightGBM	0.658	0.921	0.470
May-Jul	Weather+Dev+Soil	CatBoost	0.669	0.962	0.423
May-Aug	Weather+Dev+Soil	LightGBM	0.621	0.975	0.407
May-Sep	Weather+Dev+Soil	LightGBM	0.621	0.946	0.441
May-Oct	Weather+Dev+Soil	Ridge	0.536	0.889	0.507
May-Jun	Operational	ElasticNet	0.545	0.721	0.676
May-Jul	Operational	ElasticNet	0.515	0.702	0.693
May-Aug	Operational	ElasticNet	0.523	0.734	0.664
May-Sep	Operational	ElasticNet	0.478	0.704	0.690
May-Oct	Operational	Ridge	0.424	0.658	0.730

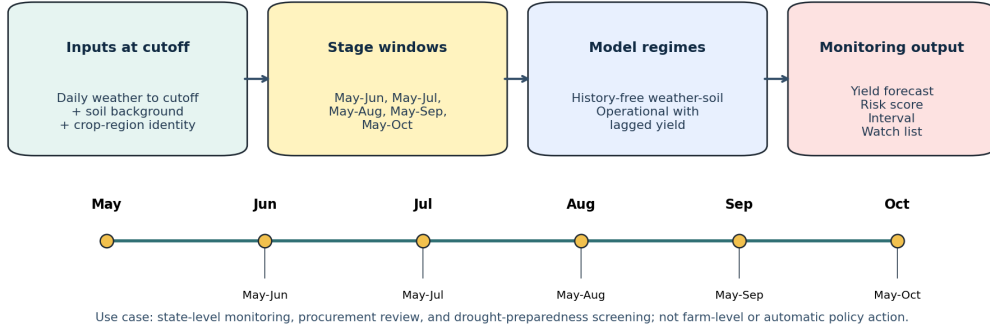


Figure 1: Stage-aware early-warning workflow. Each forecast window uses weather only up to its cutoff month, combines it with static soil and crop-region context, and updates forecasts, risk scores, intervals, and analyst watch lists.

uniformly beneficial once raw stage summaries and soil are included, which suggests that train-derived deviations should be treated as candidate monitoring features rather than guaranteed improvements. The lag-yield-only feature set is also strong, indicating that persistent crop-region yield history carries important predictive information.

Table 5 compares the May-Oct models with simple historical baselines. Previous-year yield and rolling past yield are strong, confirming that persistence matters. The operational model still improves on these baselines, while the history-free model remains useful as evidence for within-season weather-soil monitoring rather than as the best deployment model. Simple chronological yield-history baselines are therefore difficult to beat in this state-level panel. The history-free weather-soil model is not presented as the most accu-

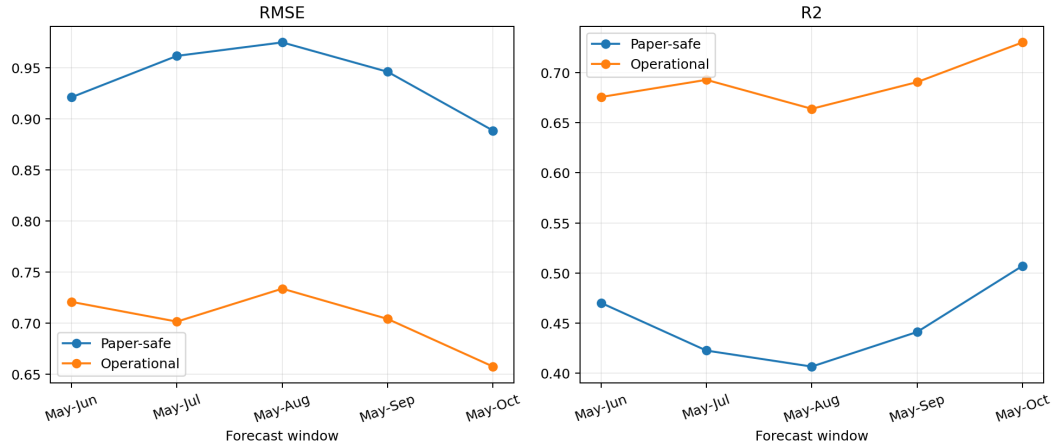


Figure 2: History-free weather-soil model versus operational model across forecast windows.

Table 4: Ablation summary for early and full-season windows. Operational models include lagged yield history; history-free weather-soil models do not.

Feature set	May-Jun RMSE	May-Jun R2	May-Oct RMSE	May-Oct R2
Identity+Time	0.856	0.543	0.856	0.543
Weather	0.920	0.472	0.875	0.522
Weather+Dev	0.931	0.459	0.889	0.507
Weather+Soil	0.913	0.480	0.865	0.533
Weather+Dev+Soil	0.921	0.470	0.889	0.507
Yield History	0.800	0.600	0.800	0.600
Operational	0.721	0.676	0.658	0.730

rate deployment option; its role is to isolate the information available from within-season weather and regional soil context, whereas the operational model is preferred for known crop-region histories.

Table 5: Naive historical baselines and May-Oct model comparison on held-out test years.

Model	Information rule	May-Oct RMSE	May-Oct R2
3-year rolling past mean	past-only chronological	0.779	0.622
Previous-year yield	past-only chronological	0.784	0.616
Crop-region train trend	train-only	0.823	0.577
Crop-region train mean	train-only	0.961	0.423
History-free weather-soil	validation-selected Ridge	0.889	0.507
Operational model	validation-selected Ridge	0.658	0.730

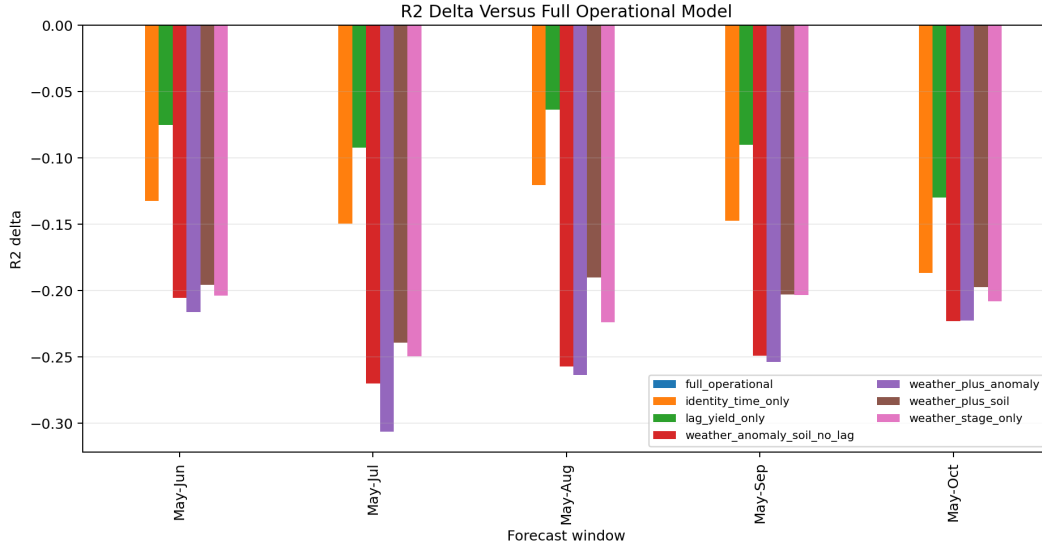


Figure 3: R2 differences relative to the full operational feature set.

6.3. Low-Yield-Risk Classification

Risk classification is treated as a decision-support layer rather than the primary performance claim. With thresholds chosen on validation years only, the best May-Oct classifier is LogisticRegression with threshold 0.27, precision 0.469, recall 0.556, F1 0.508, ROC-AUC 0.797, and PR-AUC 0.390 on the test period. The tuned threshold improves recall relative to a naive 0.5 cutoff, which is desirable for early warning where missing a shortfall may be costly. However, the modest precision means alerts should be interpreted as screening signals rather than deterministic event calls. In practice, the classifier would be most useful as a ranked watch list for analyst review, not as an automatic policy trigger.

Table 6: May-Oct low-yield-risk confusion matrix using a validation-selected threshold.

Window	Model	Thr.	TN	FP	FN	TP	Prec.	Recall	F1
May-Oct	LogisticRegression	0.27	105	17	12	15	0.469	0.556	0.508

6.4. Stress Validation

Table 8 summarizes robustness checks using the best model within each validation protocol and feature regime. Values are averaged across the folds and forecast-window rows belonging to each protocol, so the held-out-test row is a stress-validation summary rather than a duplicate of the single May-Oct result in Table 3. Rolling-origin performance remains strong, especially for the operational model, suggesting that the headline time-split result is not purely an artifact of one train-test boundary. Leave-one-crop-out is substantially harder for the history-free model, which is expected because crop identity changes the yield scale and sensitivity to weather. Leave-one-region-out produces weak or negative

Table 7: Held-out May-Oct ranked watch-list example. Risk scores are used for ranking analyst-review watch lists and should not be interpreted as fully calibrated event probabilities. The constant PI width reflects the compact conformal interval summary used for this example, not event-specific uncertainty calibration. Observed indicates the held-out low-yield-risk label.

Rank	Region	Crop	Year	Risk score	Alert	Observed	Shortfall	PI width
1	VIC	Oats	2018	1.00	1	1	0.465	1.125
2	NSW	Lupins	2018	1.00	1	1	0.410	1.125
3	VIC	Lupins	2018	1.00	1	0	0.274	1.125
4	VIC	Canola	2018	1.00	1	0	0.012	1.125
5	NSW	Wheat	2018	0.27	1	1	0.979	1.125
6	QLD	Wheat	2019	0.27	1	1	0.957	1.125
7	QLD	Wheat	2018	0.27	1	1	0.880	1.125
8	QLD	Barley	2019	0.27	1	1	0.865	1.125
9	TAS	Oats	2019	0.27	1	1	0.730	1.125
10	QLD	Canola	2018	0.27	1	1	0.646	1.125

R2. The region-transfer result is the most important limitation: the current framework is better supported for known state-level histories than for unseen-state transfer. This weakness is also consistent with the feature-importance results, where crop-region history and categorical/time structure carry substantial signal.

Table 8: Compact stress validation summary. Rows report mean performance across the folds and forecast windows belonging to each protocol for the best model within a feature regime. The held-out-test row is therefore an all-window stress summary, not the May-Oct-only result in Table 3; leave-one-region-out is a deliberately hard unseen-region test.

Protocol	Feature set	Model	Mean RMSE	Mean R2	Folds
Leave-one-crop-out	Operational	Ridge	0.731	0.520	5
Leave-one-crop-out	Weather+Dev+Soil	Ridge	1.071	0.067	5
Leave-one-region-out	Operational	Ridge	0.823	-0.356	6
Leave-one-region-out	Weather+Dev+Soil	LightGBM	0.974	-0.285	6
Rolling-origin	Operational	Ridge	0.516	0.698	3
Rolling-origin	Weather+Dev+Soil	LightGBM	0.578	0.631	3
Held-out test years, all-window mean	Operational	Ridge	0.716	0.679	1
Held-out test years, all-window mean	Weather+Dev+Soil	LightGBM	0.963	0.420	1

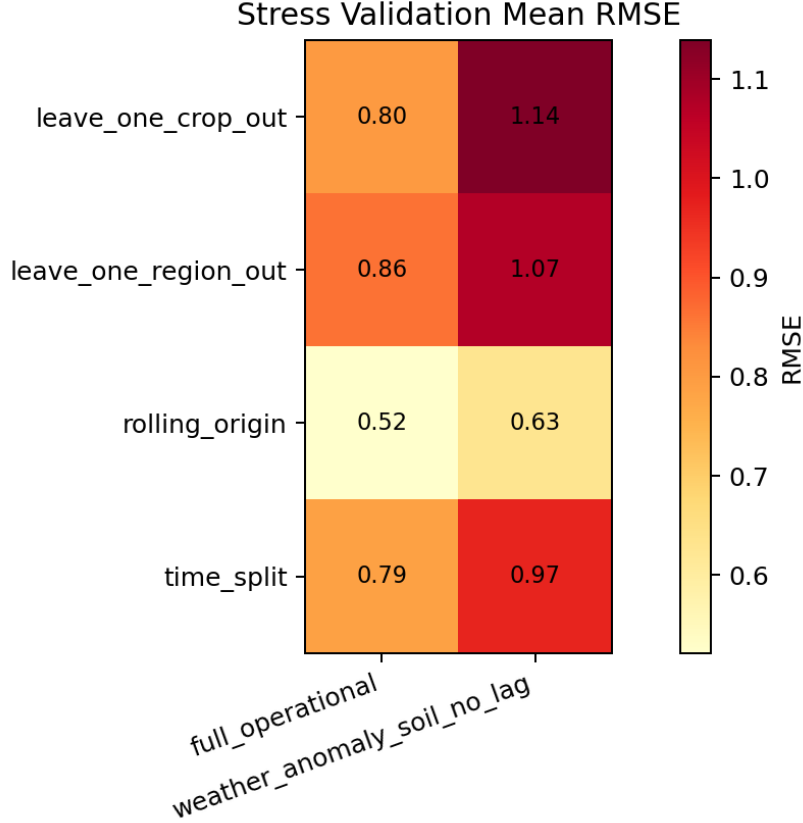


Figure 4: Stress validation heatmap by validation protocol and feature regime. Values are mean RMSE across stress-validation rows, not single-window May-Oct scores.

6.5. Feature Group Importance

Group permutation importance confirms the ablation results. Lag-yield features dominate the operational model, especially in early windows where less current-season weather has accumulated. In the history-free setting, soil background, categorical/time structure, and heat/cold features are among the most important groups. Rainfall and dry-spell features contribute in some windows, but their importance is less stable than the combined history signal. We interpret these patterns as evidence for monitoring and vulnerability conditioning rather than as causal attribution: the model can identify useful statistical signals, but the design is not an intervention study.

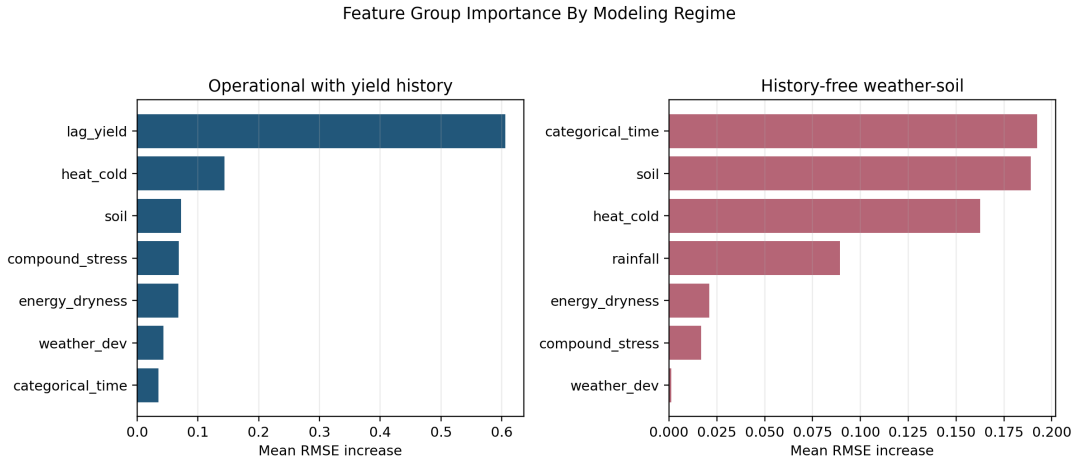


Figure 5: Feature group importance measured as mean RMSE increase after group permutation. Operational importance should not be interpreted as within-season weather importance because lag-yield features are allowed in that regime.

7. Discussion

The results suggest a two-tier interpretation. The history-free weather-soil model establishes that stage-aware weather and soil information contain measurable but secondary early-warning signal. The operational model, however, is the more accurate forecasting tool because it also uses historical yield memory. This distinction is practically important: agencies or supply-chain users with stable regional histories may prefer the operational model, while scientific analyses of within-season climate signal should emphasize the history-free model.

Taken together, the results answer the four research questions as follows. RQ1: useful operational skill is available from May-Jun and improves toward May-Oct, while history-free skill is positive but noisier. RQ2: lagged yield history and crop-region persistence explain much of the strongest accuracy, whereas weather-soil features provide a weaker but interpretable monitoring signal. RQ3: risk classification is useful as a ranked screening layer, not as an automatic trigger. RQ4: rolling-origin checks are encouraging, but leave-one-region-out transfer remains weak, limiting deployment to known crop-region histories.

The May-Jun operational result indicates that useful signal exists before the full growing season is observed. May-Oct provides the strongest final pre-harvest performance. Thus, the framework is best seen as a monitoring trajectory rather than a single fixed-date predictor. A decision maker could inspect May-Jun results as an early watch list, update the same crop-region panel as the season progresses, and use May-Sep or May-Oct results for more confident supply-risk assessment.

The strongest practical use cases are state-level monitoring, drought preparedness, procurement planning, and supply-risk assessment. The model does not decide policy or allocate payments; instead, it helps prioritize which crop-region-year combinations deserve attention. This is also why calibration, threshold tuning, and interval coverage are in-

cluded as supplementary diagnostics. Even when a point forecast is accurate on average, risk communication requires knowing how often alerts are missed, how often intervals cover outcomes, and how much uncertainty remains.

The Round 3 residual check further clarifies the claim. When a crop-region-year baseline is fitted first, a residual-only weather-soil model does not beat the direct history-free weather-soil model. This negative result supports the conservative narrative: weather and soil signals are useful, but much state-level predictability is tied to persistent crop-region structure and historical yield memory.

Soil attributes should be interpreted as regional conditioning variables. Their incremental accuracy gains are small and not uniform across feature regimes, but they help represent background vulnerability differences that would otherwise be absorbed entirely by crop-region identifiers.

8. Limitations

The analysis is state-level, not farm-level. It does not support field-scale prescription, insurance payout decisions, or causal claims about heat, drought, or soil mechanisms. The soil features are regional background covariates and should not be interpreted as independently manipulated causal factors. Leave-one-region-out validation is weak, so transfer to unseen states remains an open problem. Crop-specific phenological calendars are also left for future work; this study uses a common May-Oct winter-crop proxy for all crops. Finally, the yield panel is small by modern machine-learning standards, so future work should test whether remote sensing, crop calendars, and finer spatial labels can improve transfer without introducing leakage.

9. Conclusion

We presented a stage-aware early-warning framework for Australian winter crop yield shortfall. The best operational model reached RMSE 0.658 t/ha and R^2 0.730 at May-Oct, while the best history-free weather-soil model reached RMSE 0.889 t/ha and R^2 0.507. The gap between these regimes shows that weather and soil provide measurable but secondary monitoring signal, while historical yield memory is central to the strongest operational forecasts. This supports cautious state-level yield-risk monitoring while clearly bounding farm-level, causal, and unseen-region claims. The framework should be recalibrated before use in unseen regions or states.

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Appendix A. Supplementary Fixed-Model and Target Checks

Table 9: Fixed-model lead-time comparison. HF denotes the history-free weather-soil regime; Op. denotes the operational regime. Values are test RMSE.

Window	Ridge HF	Ridge Op.	LGBM HF	LGBM Op.
May-Jun	1.012	0.747	0.921	0.861
May-Jul	1.001	0.720	1.057	0.859
May-Aug	1.018	0.750	0.975	0.882
May-Sep	0.965	0.705	0.946	0.885
May-Oct	0.889	0.658	0.913	0.825

Table 10: Sensitivity of the low-yield-risk label to crop-specific training shortfall percentiles. The 80th percentile is the main setting.

Percentile	Train rate (%)	Validation rate (%)	Test rate (%)
70	30.1	13.8	24.2
80	20.3	11.2	18.1
90	10.4	2.6	9.4

Appendix B. Supplementary Uncertainty Diagnostics

Table 11: Compact uncertainty and calibration diagnostics. Conformal intervals target nominal 80% marginal coverage using validation-period residuals only; Brier score is from the best validation-threshold classifier for each window.

Window	Interval model	Nominal	Coverage	Width	qhat	Brier
May-Jun	Conformal ElasticNet	0.800	0.799	1.565	0.782	0.149
May-Jul	Conformal ElasticNet	0.800	0.792	1.423	0.711	0.144
May-Aug	Conformal ElasticNet	0.800	0.711	1.259	0.629	0.147
May-Sep	Conformal ElasticNet	0.800	0.758	1.251	0.626	0.197
May-Oct	Conformal ElasticNet	0.800	0.752	1.125	0.563	0.135

Detailed classification, residual-check, precision-recall, threshold trade-off, and interval coverage artifacts are retained in the reproducible project outputs for supplementary use.